Grid-scale battery dispatch distortions due to manufacturer warranties

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Motivation

- One of the supposed advantages of grid batteries is being able to quickly and flexibly respond to the grid compared to other resources
- However, performance warranties set by the battery's manufacturer often include restrictions on the battery's dispatch behavior in order for the warranty to remain valid.

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Research question: How do cycling behavior restrictions in grid-scale battery warranties distort battery behavior away from profit-maximizing (or socially optimal) dispatch?

Context: Grid-scale battery warranties

- **Commercial Warranties**: a short-term guarantee that covers defects, part replacements, and component failures at the beginning of life
- **Performance Warranties**: guarantee that the storage system will maintain specific performance standards over a longer period—often between 15-20 years

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Context: Types of Performance Warranties

- **End-of-Life State of Health (SoH) warranty**: Promises a specific SoH (typically 70–80%) by the end of the warranty period if the battery is used within certain operating conditions.
- **Annual SoH warranty**: Promises a maximum yearly degradation rate if the battery is used within certain operating conditions.
- **Penalty-based warranty**: Allows for adjustments in the yearly SoH guarantee depending on how battery is operated (e.g. if the system operates under sub-optimal conditions, the guaranteed SoH is reduced by a predefined formula.)

Context: Types of operating restrictions

Restrictions imposed on battery operation to prevent degradation:

- Limiting the number of cycles per year or per day
- State-of-charge requirements (e.g. less than 50% annual average SOC)
- Depth-of discharge limitations
- Resting period requirements
- etc. etc. etc.

Context: Trends in battery warranties

Generally, the industry has been interested in moving away from more **restrictive** warranties (e.g.limits on the number of cycles) to more **flexible** warranties (e.g. rolling averages, charge/discharge rates over the day, etc).

Reasons for change:

- Batteries mainly provide ancillary services, so applying a restriction on full cycles makes less sense
- Batteries make much of their money arbitraging on a few high-demand days, rather than just once every days, so restrictions can prevent optimal dispatch.
  - *"Running a BESS with an inflexible guarantee is like paying rent all year for an umbrella shop, but being forced to keep the same daily stock; so when the rainy week comes, you sell out in the first hour and miss the payoff"* *-Gregory Green, managing partner at BlackVolt.*

Example: Trina Storage

- Product of Trinasolar, a Chinese PV/storage manufacturer
- **Early, pre-2025 warranties**: allows for 4 full-power cycles in a day
- **New 2025 warranties**: limits continuous operation to around 0.25C charge/discharge rate (power/capacity)

Literature

- **Electricity storage arbitrage and optimal dispatch**: Profit-maximizing battery operation in electricity markets
  Sioshansi et al. (2009); Sioshansi (2010); Denholm and Margolis (2018)
- **Contract design and warranties**: How contracts and warranties restrict buyer actions to manage risk, particularly for cars
  Cooper and Ross (1985); Emons (1989); Murthy and Ye (2016)
- **Optimal design of battery warranties**
  Liu and Wang (2025)

Contribution: Unlike cars and many other products, for a battery, 1) there are several different dimensions of operational behavior a manufacturer could contract on, and 2) these choices could have larger effects on the the electricity market.

Data sources

Data on **operation restrictions in warranties**:

- (in-progress) LBNL has a sample of 105 PV+battery hybrid plants with public PPAs, some of which contain restrictions on battery operations such as number of cycles per year or day, state-of-charge requirements, etc due to battery warranties.
- (in-progress) Reach out to companies specializing in constructing adaptive energy storage warranties (NEC Energy Solutions, IronGrid) to see if they are willing to provide information on battery manufacturers who switched to more flexible warranties.

Data on **battery operation**:

- ERCOT's 60-day SCED Disclosure Report

Research Plan

- Categorize batteries into buckets (more vs less restrictive warranties), depending on observed variation among PPAs
- Non-causal descriptives: compare battery behavior, such as 1) their bids into the ancillary services market and 2) their bids in response to unexpected price dips / spikes
- Potential simulation:
  - Give each type of battery (+ a theoretical "no restriction" battery) the same price forecasting software, based on weather and previous prices.
  - Simulate their optimal behavior for an empirical price series for a year.
  - Calculate difference in battery profit.
  - Compare with degradation costs for that dispatch behavior (to take from engineering literature / modeling)